

CHAPTER 1

INTRODUCTION TO THE POWERED PARACHUTE

History of the Powered Parachute

As early as the 12th century, the Chinese used an umbrella-shape parachute design for recreation. About 300 years later, Leonardo da Vinci blueprinted a pyramid-shaped parachute. In the late 18th century, man jumped from towers and balloons with a parachute. The first parachute jump from an airplane occurred in 1912.

After World War II, sport jumping became a recreational activity. The sport started with round parachutes, ranging in size from 20 to 30 feet in diameter. Parachutes evolved into a steerable, gliding wing smaller than today's rectangular ram-air powered parachute (PPC) wing which is approximately 38 feet wide.

On October 1, 1964, Domina C. Jalbert applied for a patent for his "Multi-Cell Wing" named "Parafoil" (also known as a "ram-air" wing), which was a new parachute design. His ideas were registered as a U.S. patent on November 15, 1966. [Figure 1-1 A] However,

in 1964 Lowell Farrand had already flown a motorized version called "The Irish Flyer" by Nicolaides. [Figure 1-1 B] Farrand was the first person to put an engine on a ram-air inflated parachute wing, starting the evolution of the powered parachute with the Irish Flyer. This wing evolved into today's modern powered parachute canopies, which include rectangular, elliptical, semi-elliptical, and hybrid wings.

The United States (U.S.) government had a number of test programs that used the square parachute as a means to glide spacecraft back to earth or glide payloads dropped out of airplanes to a specific location.

Two-place powered parachutes have years of testing, development, and evolution. Training exemptions to Title 14 of the Code of Federal Regulations (14 CFR) part 103, Ultralight Vehicles, permitted individuals to give instruction in two-place ultralight vehicles, instead of being restricted to vehicles intended for single occupants. [Figure 1-1 C] The Federal Aviation Administration (FAA) allowed ultralight vehicle

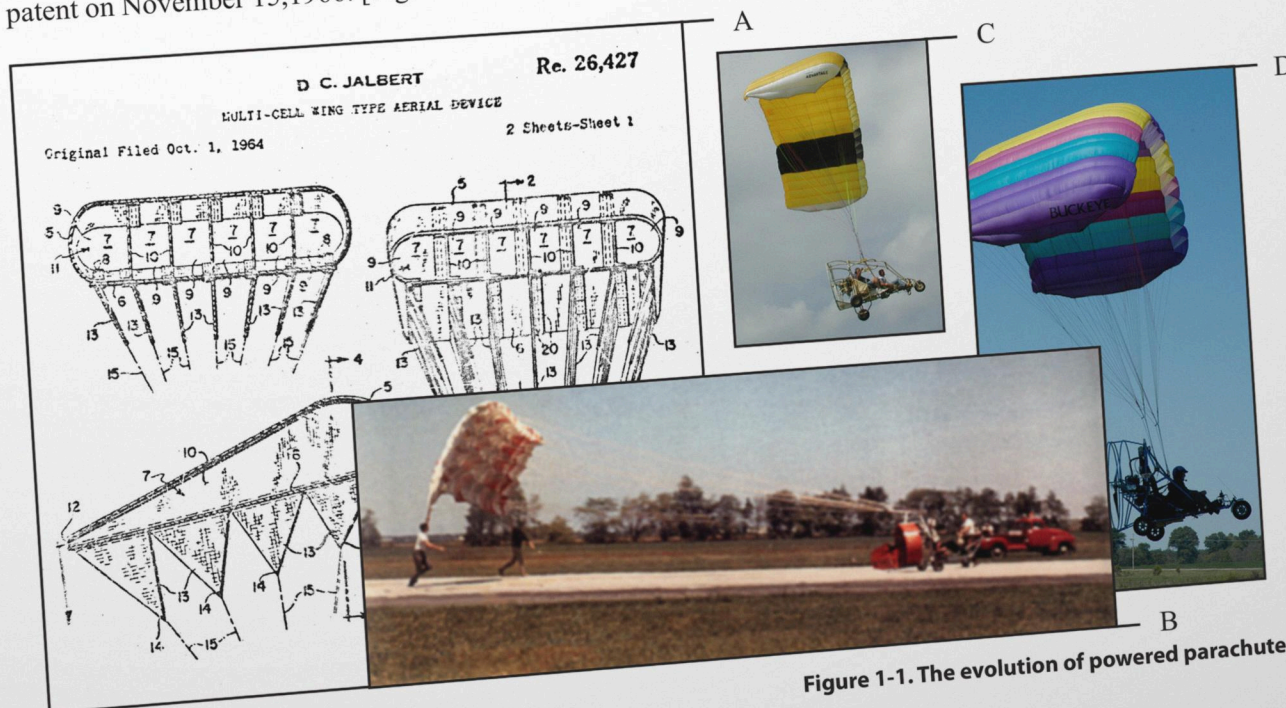


Figure 1-1. The evolution of powered parachutes.